

Option-Implied Probabilities

BayQuant

Options as Bets on States

A call struck at K pays only in the states where the underlying ends above K . Calls at two nearby strikes differ only in the states between them, so the market's prices across strikes say how much it charges for each range of outcomes. Breeden and Litzenberger [1] made this exact: the second strike derivative of the call price is the discounted risk-neutral density of the underlying at expiry. Ross [2] had shown that options on a single asset span its states; Breeden and Litzenberger turned that into a formula.

This note derives the result, checks it numerically under Black–Scholes [3] and under a bimodal event distribution, shows how the implied-volatility smile enters event probabilities, and explains why the formula cannot be applied to raw quotes.

State Prices

Let $q(S)$ be the risk-neutral density of S_T , and r the continuously compounded rate. Any European payoff is worth its discounted risk-neutral expectation, so

$$C(K) = e^{-rT} \int_K^\infty (S - K)q(S) dS. \quad (1)$$

The strike appears twice: in the lower limit and inside the integrand. Differentiating such an integral takes the Leibniz rule. For a lower limit $a(K)$,

$$\begin{aligned} \frac{d}{dK} \int_{a(K)}^\infty f(S, K) dS &= -f(a(K), K)a'(K) \\ &\quad + \int_{a(K)}^\infty \frac{\partial f}{\partial K} dS. \end{aligned}$$

The first term comes from moving the limit, the second from changing the integrand.

First derivative. Here $a(K) = K$, so $a'(K) = 1$, and $f(S, K) = (S - K)q(S)$. The boundary term is $-(K - K)q(K) = 0$: the payoff is zero at the strike, so moving the strike loses nothing there. Inside the integral, $\partial f / \partial K = -q(S)$. So

$$\frac{\partial C}{\partial K} = -e^{-rT} \int_K^\infty q(S) dS.$$

The remaining integral is the area under the density to the right of K , which is the probability of finishing above K :

$$\frac{\partial C}{\partial K} = -e^{-rT} \mathbb{Q}(S_T > K). \quad (2)$$

Second derivative. Apply the Leibniz rule to the integral in the previous step. Now $f(S, K) = q(S)$ does not depend on K , so the integrand term vanishes and only the boundary term, $-q(K)$, is left:

$$\frac{\partial^2 C}{\partial K^2} = -e^{-rT} \cdot (-q(K)) = e^{-rT} q(K). \quad (3)$$

The slope of the call curve is a discounted digital option, and its curvature is a discounted density: the price today of \$1 paid if S_T lands in $[K, K + dK]$, per unit dK . These are the Arrow–Debreu state prices. Nothing is assumed about the dynamics of S ; the result needs only calls at every strike. Puts give the same second derivative (appendix).

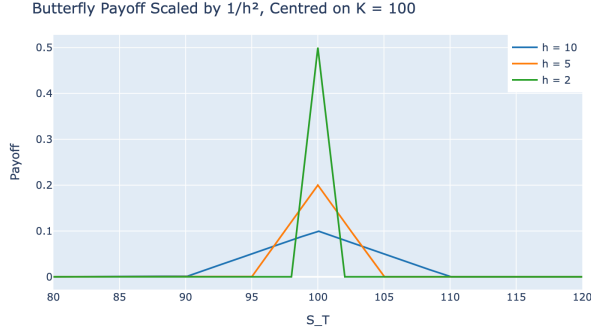
Two no-arbitrage conditions follow for free. Because $q \geq 0$, the call curve is convex in K . Because $\mathbb{Q}(S_T > K)$ lies in $[0, 1]$, its slope lies in $[-e^{-rT}, 0]$. Any fitted call curve that breaks either condition implies negative probabilities.

The Butterfly

The second difference

$$\frac{C(K - h) - 2C(K) + C(K + h))}{h^2} \rightarrow \frac{\partial^2 C}{\partial K^2} \quad (4)$$

is the price of a portfolio: long one call at $K - h$, short two at K , long one at $K + h$, scaled by $1/h^2$. Its payoff is a triangle of height $1/h$ and area 1 centred on K . As h shrinks it narrows into a spike, the payoff of an Arrow–Debreu security. The butterfly is what makes state prices tradable.



Probabilities of Events

Equation (2) gives event probabilities without any density or integration:

$$\mathbb{Q}(S_T < K) = 1 + e^{rT} \frac{\partial C}{\partial K}. \quad (5)$$

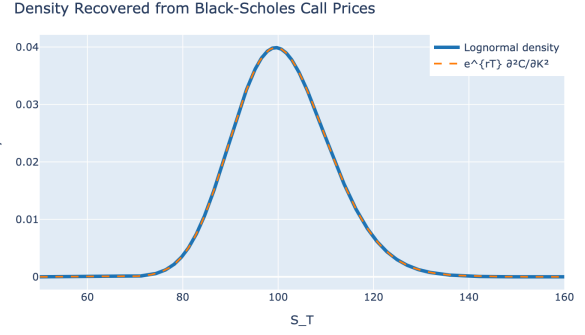
The traded version is a tight call spread, long $K - h/2$ and short $K + h/2$, scaled by $1/h$: a digital paying \$1 above K . Under Black–Scholes, with forward $F = S_0 e^{rT}$,

$$\mathbb{Q}(S_T > K) = N(d_2), \quad q(K) = \frac{n(d_2)}{K\sigma\sqrt{T}}, \quad (6)$$

where $d_2 = (\ln(F/K) - \sigma^2 T/2)/\sigma\sqrt{T}$. The probability of finishing in the money is $N(d_2)$, not the delta $N(d_1)$.

Check Under Black–Scholes

Take a \$100 stock, three months to expiry, $r = 4\%$ and $\sigma = 20\%$. Pricing calls with Black–Scholes and applying (4), the recovered density matches (6). The central difference has error $h^2/12$ times the density's second derivative (appendix), so halving h should cut the worst-case error fourfold. It does: 5.3×10^{-4} , 1.4×10^{-4} , 3.4×10^{-5} and 8.5×10^{-6} for $h = 4, 2, 1, 0.5$. On a grid 5 cents apart the recovered density integrates to 1.0000 and its mean is the forward, 101.00. Call spreads reproduce $N(d_2)$ to four decimals, giving $\mathbb{Q}(S_T < 90) = 13.5\%$ and $\mathbb{Q}(S_T > 110) = 18.3\%$.



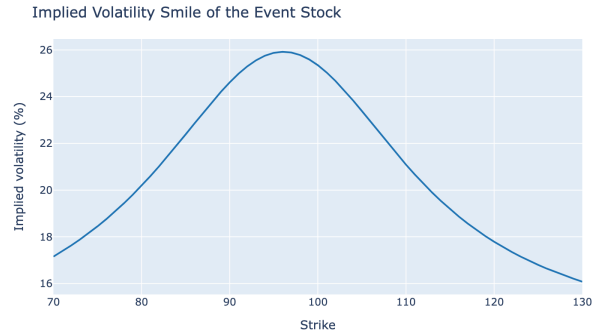
A Scheduled Binary Event

The result is model-free, so it must recover any distribution, including ones no smooth model would produce. Consider a stock facing a binary event before expiry, such as a regulatory decision. With probability $w = 0.6$ it ends near $F_{\text{up}} = 1.08F = 109.09$, otherwise near $F_{\text{down}} = 88.88$, with volatility $s = 12\%$ within each scenario. F_{down} is set so that $wF_{\text{up}} + (1-w)F_{\text{down}} = F$, as risk-neutral pricing requires. A mixture of lognormals prices calls as the same mixture of Black–Scholes prices,

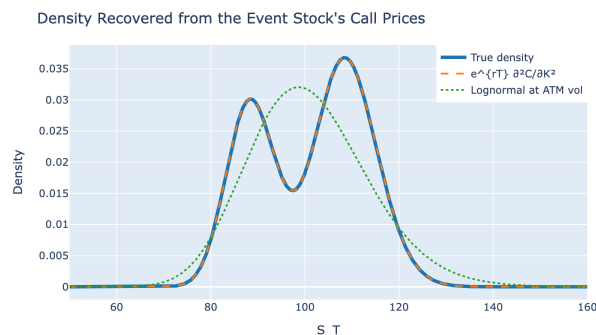
$$C(K) = wC_{\text{BS}}(F_{\text{up}}, K, s) + (1-w)C_{\text{BS}}(F_{\text{down}}, K, s), \quad (7)$$

the approach Melick and Thomas [4] used for crude oil during the Gulf crisis. Here it is the truth, so the recovery can be checked.

A trader quoting these options in Black–Scholes terms sees each price as its own implied volatility, and the curve is an inverted smile. It peaks at 26% near $K = 96$, between the two outcomes, where the event decides whether the option pays, and falls to about 16% in either wing, where the outcome barely matters.



Applying (4) to the prices recovers both humps, with a maximum error of 2×10^{-7} . A lognormal at the at-the-money volatility puts its peak near the trough between the two outcomes.



The Smile Moves Probability

A tempting shortcut is to read $\mathbb{Q}(S_T > K)$ as $N(d_2)$ evaluated at the implied volatility $\sigma(K)$ of each strike. With a smile that is wrong. Writing $C(K) = C_{BS}(K, \sigma(K))$ and differentiating,

$$\mathbb{Q}(S_T > K) = N(d_2(\sigma(K))) - e^{rT} \mathcal{V}(K) \sigma'(K), \quad (8)$$

where $\mathcal{V}$ is the Black–Scholes vega. The slope of the smile shifts probability between strikes. For the event stock:

K	true	$N(d_2)$	with (8)
85	0.9050	0.9313	0.9050
95	0.6441	0.6588	0.6441
105	0.4377	0.3482	0.4377
115	0.1088	0.0806	0.1088

At $K = 105$, where the curve falls steeply, the shortcut understates the chance of the good outcome by 9 points. With the skew term, (8) is exact.

Why the Recipe Fails on Quotes

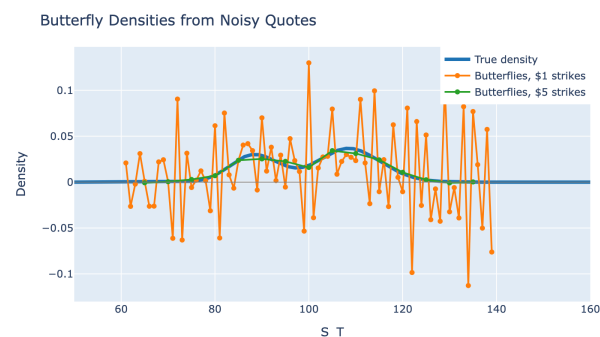
Everything so far used exact prices on a grid 5 cents apart. Real options list at a few strikes, often \$1 to \$5 apart, and each quote is known only to within its bid–ask spread. Both damage the second difference (appendix):

- spacing h adds a bias $h^2 q''(K)/12$, which smears sharp features such as the event’s two humps;
- independent quote errors of size ε add density noise of size

$$e^{rT} \sqrt{6\varepsilon}/h^2. \quad (9)$$

Shrinking h cuts the bias and inflates the noise. With two cents of quote error, far inside a typical spread, the noise is 0.050 at \$1 spacing, larger than the peak density of 0.037, and 29 of 79 butterflies imply negative

probabilities. At \$5 spacing the noise is 0.002, but the two humps blur together and nothing is known between strikes or beyond the last one.



Neighbouring butterflies share quotes: a price that is too high raises the butterfly centred on it and lowers both neighbours. The noise therefore alternates in sign from strike to strike, the zigzag in the figure.

What Comes Next

The theory is exact, but it needs a smooth, arbitrage-free call curve at every strike, and markets supply a few noisy points. The practical methods differ in how they fill the gap. Shimko [5] smooths the implied-volatility curve and differentiates the call prices it implies. Jackwerth and Rubinstein [6] solve for the smoothest density that prices the quotes within their spreads, which makes the fitted call curve arbitrage-free by construction.

Whatever the method, the result is a risk-neutral distribution. It prices outcomes in dollars today, weighted by how much investors fear them, and generally puts more mass on bad states than their real-world frequency.

Questions

1. Why is the probability written $\mathbb{Q}$?

$\mathbb{Q}$ marks the *risk-neutral* probability, to keep it apart from the real-world probability $\mathbb{P}$. The convention comes from Harrison and Kreps [7], who showed that absence of arbitrage is equivalent to the existence of a probability measure, called $\mathbb{Q}$, under which every price is a discounted expectation:

$$\text{price} = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\text{payoff}].$$

This is the rule used in (1). The idea goes back to Cox and Ross [8]: since an option's price does not depend on investors' risk preferences, it can be computed as if everyone were risk-neutral. Under $\mathbb{Q}$ every asset is expected to grow at the risk-free rate, so $\mathbb{E}^{\mathbb{Q}}[S_T] = F$. That is why the density recovered under Black-Scholes has mean 101.00, the forward, and not whatever return the stock is really expected to earn.

How $\mathbb{Q}$ differs from $\mathbb{P}$. The price of \$1 paid in a state depends on two things: how likely the state is, and how much investors value a dollar in it. With $p(S)$ the real-world density of S_T and $m(S)$ the value of a dollar in that state (the stochastic discount factor, or pricing kernel), the state price is $p(S)m(S)$. By (3) it is also $e^{-rT}q(S)$. Integrating both over all states fixes $e^{-rT} = \mathbb{E}^{\mathbb{P}}[m]$, so

$$q(S) = \frac{m(S)}{\mathbb{E}^{\mathbb{P}}[m]} p(S).$$

$\mathbb{Q}$ reweights $\mathbb{P}$ by how much each state matters to investors. A dollar is worth most in a crash, when it is scarce, so m is high in bad states and $\mathbb{Q}$ puts more weight there than $\mathbb{P}$ does. The two agree only if investors are risk-neutral, with m constant.

What this means for the numbers. $\mathbb{Q}(S_T < 90) = 13.5\%$ under Black-Scholes is a price, not a forecast: a claim paying \$1 if $S_T < 90$ costs $e^{-rT} \times 0.135 = 0.134$ dollars today. To turn it into a real-world probability one needs m , which option prices alone do not reveal without extra assumptions. Ross [9] gives conditions under which they do.

Appendix

State Prices

Equations (2) and (3)

Differentiate (1) with the Leibniz rule. The lower limit contributes $-(K - K)q(K) = 0$, and differentiating the integrand gives $-q(S)$, so

$$\frac{\partial C}{\partial K} = -e^{-rT} \int_K^\infty q(S) dS = -e^{-rT} \mathbb{Q}(S_T > K).$$

Differentiating again, the integral loses its lower limit:

$$\frac{\partial^2 C}{\partial K^2} = e^{-rT} q(K).$$

For a put, $P(K) = e^{-rT} \int_0^K (K - S)q(S) dS$ gives $\partial P/\partial K = e^{-rT} \mathbb{Q}(S_T < K)$. Put-call parity, $C - P = e^{-rT}(F - K)$, is linear in K , so C and P have the same second derivative.

Puts

A put pays $K - S$ in the states below the strike:

$$P(K) = e^{-rT} \int_0^K (K - S)q(S) dS.$$

Now K is the upper limit, and the Leibniz rule for an upper limit $b(K)$ adds $+f(b(K), K)b'(K)$ instead of subtracting. The boundary term is again $(K - K)q(K) = 0$, and $\partial f/\partial K = q(S)$, so

$$\frac{\partial P}{\partial K} = e^{-rT} \int_0^K q(S) dS = e^{-rT} \mathbb{Q}(S_T < K).$$

Differentiating once more leaves only the boundary term, $+q(K)$:

$$\frac{\partial^2 P}{\partial K^2} = e^{-rT} q(K).$$

This matches (3). It also follows from put-call parity, $C - P = e^{-rT}(F - K)$: the difference is linear in K , so C and P have the same second derivative.

Black-Scholes

Equation (6)

On the forward, $C = e^{-rT}(FN(d_1) - KN(d_2))$ with $d_1 = d_2 + \sigma\sqrt{T}$. Two facts simplify the derivative. First, $\partial d_1/\partial K = \partial d_2/\partial K = -1/(K\sigma\sqrt{T})$. Second, $Fn(d_1) = Kn(d_2)$, since

$$\ln \frac{n(d_1)}{n(d_2)} = -\frac{d_1^2 - d_2^2}{2} = -(d_1 - d_2)\frac{d_1 + d_2}{2} = -\ln(F/K).$$

Differentiating C ,

$$\frac{\partial C}{\partial K} = e^{-rT} \left[Fn(d_1) \frac{\partial d_1}{\partial K} - N(d_2) - Kn(d_2) \frac{\partial d_2}{\partial K} \right] = -e^{-rT} N(d_2),$$

because the first and third terms cancel. Differentiating again,

$$\frac{\partial^2 C}{\partial K^2} = -e^{-rT} n(d_2) \frac{\partial d_2}{\partial K} = e^{-rT} \frac{n(d_2)}{K\sigma\sqrt{T}}.$$

Comparing with (2) and (3) gives (6).

Smile Correction

Equation (8)

When each strike has its own implied volatility, $C(K) = C_{\text{BS}}(K, \sigma(K))$ and the chain rule gives

$$\frac{dC}{dK} = \frac{\partial C_{\text{BS}}}{\partial K} + \frac{\partial C_{\text{BS}}}{\partial \sigma} \sigma'(K) = -e^{-rT} N(d_2) + \mathcal{V} \sigma'(K),$$

with vega $\mathcal{V} = e^{-rT} F n(d_1) \sqrt{T}$. Multiplying by $-e^{rT}$ and using (2) gives (8). Where the smile slopes down ($\sigma' < 0$), the correction raises $\mathbb{Q}(S_T > K)$ above $N(d_2)$.

Discretization Error and Noise

Equation (9)

Expanding $C(K \pm h)$ in a Taylor series, the odd terms cancel in the second difference:

$$\frac{C(K-h) - 2C(K) + C(K+h)}{h^2} = C'' + \frac{h^2}{12} C^{(4)} + O(h^4).$$

Multiplying by e^{rT} and using (3), $e^{rT} C^{(4)} = q''$, so the recovered density is off by $h^2 q''(K)/12$.

Now let each quote carry an independent error ε_i with standard deviation ε . The butterfly combines them as $\varepsilon_{i-1} - 2\varepsilon_i + \varepsilon_{i+1}$, whose variance is $(1 + 4 + 1)\varepsilon^2$. The density error therefore has standard deviation $e^{rT} \sqrt{6}\varepsilon/h^2$. Adjacent butterflies share two quotes, and the correlation between neighbours is

$$\frac{\text{Cov}(-2\varepsilon_i + \varepsilon_{i+1}, \varepsilon_i - 2\varepsilon_{i+1})}{6\varepsilon^2} = \frac{-2 - 2}{6} = -\frac{2}{3},$$

which produces the alternating signs.

Notation

symbol	meaning
S_T	underlying price at expiry
S_0, F	spot price and forward, $F = S_0 e^{rT}$
K	strike
T	time to expiry, years
r	continuously compounded rate
$q(S)$	risk-neutral density of S_T
$\mathbb{Q}$	risk-neutral probability
$\mathbb{P}$	real-world probability
$C(K), P(K)$	call and put prices
h	strike spacing
$\sigma, \sigma(K)$	volatility; implied volatility at K
N, n	standard normal cdf and pdf
d_1, d_2	Black-Scholes terms, $d_1 = d_2 + \sigma\sqrt{T}$
$\mathcal{V}$	Black-Scholes vega
w	probability of the good event outcome
$F_{\text{up}}, F_{\text{down}}$	scenario forwards

symbol	meaning
s	volatility within each scenario
ε	standard deviation of quote error

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